

POC

Pakashona POC Specification Document (v1.0)

Project: Pakashona — Digital Gold Investment & Gamified Savings Platform

Pakashona is a **gamified digital gold savings** platform powered by **I.K. Jewellers** (Bangladesh).

Objective

Validate the functional logic and backend flows of Pakashona's digital gold investment platform before UX/UI design and mobile development.

Users can:

- **Buy** gold from the **store** (cash) or **in-app (payment gateway)**
- **View** live **gold market price**
- **Exchange** gold for jewellery at the store
- **Sell** gold to their bank accounts (within 3–5 business days), in store
- **Track Gold** progress through **gamified jewellery levels**, where grams = progress points

In-Scope

1. **User Registration & Auth**
 - Phone-based OTP login.
 - Admin/store login (email/password).
2. **Gold Price Management**
 - Admin sets the current price per gram (BDT).
 - Snapshot of price stored for each transaction.
3. **Wallet System**
 - User wallet stores balance in grams.
 - Transactions restricted to 0.5g increments.
4. **Transaction Flows**
 - Buy (in-store / in-app)
 - Exchange (store approval, –10% deduction of current price)
 - Sell to bank (2% fee, 3–5 days processing)
 - Sell to Store (17% deduction)
5. **Gamified Levels**
 - Hardcoded levels (Nosepin, Ring, Churi, Chain, etc.)
 - Live calculation of unlocked levels.
 - Badges only
6. **Admin/Store Panel**
 - View transactions awaiting approval.
 - Approve/reject codes (store exchanges).
 - Mark sells requests as “Paid.”

Transaction Flow Summary

- **Buy (Store Flow)**
 - The user buys gold in-store
 - Admin/store enters in the dashboard.
 - Backend validates, updates wallet.
 - The app reflects the updated wallet instantly.
- **Buy (App Flow)**
 - The user buys gold
 - Payment Gateway Approves
 - Backend validates, updates wallet
 - The app reflects the updated wallet instantly.
- **Exchange (Jewellery)**
 - User generates exchange codes via app.
 - Lock the grams
 - Store/admin confirms via dashboard
 - Deducts 10%, updates wallet
 - The app shows status + adjusted grams.
- **Sell (to Bank)**
 - User requests to sell X grams (multiple of 0.5).
 - Backend applies 2% total deduction, locks grams.
 - Admin reviews, marks PAID.
 - The app displays "Payment sent" with a timeline.
- **Sell (to Store)**
 - User requests to sell X grams (multiple of 0.5).
 - Backend applies 17% total deduction, locks grams.
 - Admin reviews, marks PAID.
 - The app shows status + adjusted grams.

Gamification

Balance changes → Levels update in real-time

- Levels shown live in UI (calculated each render).
- If balance drops below threshold → badge lost.
- No tangible perks in POC; just visible status.

MVP Roadmap

Pakashona MVP (v1.2)

Project: Pakashona — Digital Gold Investment & Gamified Savings Platform

Pakashona turns gold saving into a modern, gamified experience for Bangladesh. Users buy and store real, redeemable gold—unlocking jewellery “levels” as they save. Their gold can be instantly converted into jewellery on preferred designs through I.K. Jewellers, bridging digital convenience with cultural tradition. Pakashona makes heritage savings simple, visual, and trustworthy for the next generation.

CodeBase	<githublink>
Backend API Docs	: Pakashona/docs/api_endpoints.md
Architecture	: figma design
ADR	
ClickUp Board	https://app.clickup.com/90131221102/v/li/901311422019

Purpose

The goal of this MVP is to turn Pakashona’s core product logic into a clear, intuitive mobile experience that:

- Makes gold investment feel easy and secure
- Visually connects digital savings with the emotional value of jewellery
- Builds user trust through clarity, transparency, and progress visualization.
- This is not a generic finance app — it’s gold made playful, personal, and real.

What Pakashona Is

- A mobile app that lets users invest in digital gold, track their savings progress like a game to see the next unlocked gold jewellery, and exchange or sell gold directly with I.K. Jewellers.

Core Product

- Digital holding of 24K (acid gold)
- Digital holding of 22K/21K gold

Audience Trust Partner For Gold Vault/Holding

- I.K Jewellers Ltd.

Target Audience

- Young working professionals (ages 22–45)
- People who trusts in physical gold but want a digital experience
- People who want to invest long term as a form of heritage and bridal savings
- Rural areas where its hard to store gold
- Lacks trust in digital investment platforms

- Users with limited exposure to investment apps.
- People saving up for jewellery purchases, gifts, or long-term value.
- They are emotion-driven investors, not traders.
- They respond more to progress visuals than raw numbers.

Core User Goals

- See how much gold they own.
- Buy more gold (in-app or by linking store purchase).
- Sell or exchange their gold confidently.
- See live gold prices and feel in control.
- Track savings progress visually through jewellery “levels.”

Core App Experiences to Design

A. Onboarding / Trust Building

Clean, elegant welcome screens introducing Pakashona’s concept.

OTP-based sign-up with minimal friction.

Explain “Your gold is real, stored and redeemable at I.K. Jewellers.”

Ask for name, phone, and bank info in a friendly tone — not bureaucratic.

B. Dashboard (Home Screen)

This is the heart of the app — the user’s relationship with their gold.

Must show:

Wallet Balance (grams + BDT value)

Live gold price ticker (auto updates)

“Jewellery Progress” bar or ladder (nosepin → ring → chain → necklace)

Quick buttons: Buy Gold, Sell Gold, Exchange Jewellery

C.1. Buy Gold Flow 22K/21K

Two paths:

Digital Purchase (via bKash/Nagad/Card)

The user selects grams (only multiples of 0.5g).

Sees current price and small 2% platform fee.

Confirmation screen → “Your gold has been added.”

In-store Purchase (link with store QR)

User scans receipt QR → gold instantly appears in app.

Small “verified by I.K. Jewellers” label builds trust.

C.2 Buy Gold Flow 24K

Two paths:

Digital Purchase (via bKash/Nagad/Card)

The user selects grams (only multiples of 0.5g).

Sees current price and small 2% platform fee.

Confirmation screen → “Your gold has been added.”

In-store Purchase (link with store QR)

User scans receipt QR → gold instantly appears in app.
Small “verified by I.K. Jewellers’ label builds trust.

D.1 Sell Gold Flow 24K

Two paths:

Digital

User taps “Sell Gold.”
Enters grams (0.5g increments).
Sees breakdown: market rate, 2% deduction, estimated payout date.
Confirmation → success state with timeline (“3–5 business days”).

In Store

Select grams
App generates QR/code → store staff scans.
User gets real-time “Approved ” update.

D.2. Sell Gold Flow 22K/21K

Two paths:

Digital

User taps “Sell Gold.”
Enters grams (0.5g increments).
Sees breakdown: market rate, 2% deduction, estimated payout date.
Confirmation → success state with timeline (“3–5 business days”).

In Store

Selects grams → app shows “This equals X BDT after 10% jewellery rate.”
App generates QR/code → store staff scans.
User gets real-time “Approved ” update.

E. Exchange for Jewellery

User taps “Exchange.”

Selects grams → app shows “This equals X BDT after 10% jewellery rate.”
App generates QR/code → store staff scans.
User gets real-time “Approved ” update.

F. Gamified Levels (Emotional Layer)

Each jewellery type represents a savings milestone.

Levels example:

Nosepin
2.5g
First gold savings goal
Ring
5.0g
Personal milestone
Churi
10.0g

Family milestone
Chain
25.0g
Confidence milestone
Necklace
50.0g
Freedom milestone

When a user unlocks a level:

Show badge animation + motivational line (“You’ve reached Ring Level – you’re shining brighter”)

Show next goal below (“5g more for Churi Level!”)

G. Notifications

Design simple push/in-app notifications for:

Price changes (“Gold is up 1.2% today!”)

Status updates (“Your exchange has been approved.”)

Level unlocks (“You reached Ring Level!”)

Tone: warm, human — never robotic.

H. Admin/Store Interface (Optional for Now)

Simple web dashboard for store approval flows.

Focus on clarity: list of transactions, approve/reject buttons, search by user/phone.

Clean table of pending transactions

Approve/Reject buttons

Mark as paid functionality

Price setting interface

Key Design Principles

Trust first. Always show price, rate, and fees before confirming.

Visual progress. Gold = growth; show that.

Minimal input effort. Avoid long forms.

Bangladeshi context. Price in BDT, small cultural cues (ornaments, tone).

Reward emotions. Treat savings as achievements, not math.

MVP Questions

- When a wallet has locked certain grams of gold, should we allow the user to make more transactions? (currently user can perform multiple locking till total_grams)
- As we lock the grams in the backend, this will still deduct the price in the client side. So we will not be fetching directly the user balance from the backend but the (balance - lock).
- Should deduction happen **at code generation** or **upon store approval**? (Current lean: approval stage.)
- Should codes expire after a fixed time? (To prevent confusion.)
- Should admin be able to override prices post-snapshot?
- How to handle race conditions between “sell” and “exchange” if user spams actions?
-
-

Business Logic

Business Rules 24K– Gold Trading Platform (v1.4)

Action	Fee/Deduction	Approval Flow	System Behaviour/Notes
Buy in Store	8% + VAT	Admin approves	Admin manually credits purchased grams to the user wallet after physical purchase verification.
Buy in App	8% + VAT	Auto (payment gateway event)	Payment gateway collects funds → admin wallet receives → system automatically credits grams to user wallet.
Exchange (Jewellery Conversion From 22K storage)		Sell the 24K gold, put it in a table waiting for admin to convert to exchange **	When user exchanges jewellery for wallet gold, system deducts 10 % (conversion & handling) before updating balance.
Sell to Store From 24K storage	0%	Admin or store approves	Instant cash-out. Deduction includes design loss, spread, and resale cost. Funds released on-site.
Sell to Bank	0%	Admin marks 'paid' after transfer	Redemption takes 3–5 business days. User notified when payment settles.

Business Rules 22K– Gold Trading Platform (v1.3)

Action	Fee / Deduction	Approval Flow	System Behavior / Notes
Buy in Store	0 %	Admin approves	Admin manually credits purchased grams to the user wallet after physical purchase verification.
Buy in App	TBD (Payment Gateway fee)	Auto (payment gateway event)	Payment gateway collects funds → admin wallet receives → system automatically credits grams to user wallet.
Exchange (Jewellery Conversion From 22K storage)	10 % deduction	Admin or store approves	When user exchanges jewellery for wallet gold, system deducts 10 % (conversion & handling) before updating balance.
Sell to Store From 22K storage	17 % deduction	Admin or store approves	Instant cash-out. Deduction includes design loss, spread, and resale cost. Funds released on-site.
Sell to Bank	2 % fee deduction	Admin marks 'paid' after transfer	Redemption takes 3–5 business days. User notified when payment settles.
Gamified Levels	N/A (calculated live)	Auto	Levels/badges (e.g. Bronze, Silver, Gold) are computed dynamically based on wallet grams. Affects only user UI, not transactions.
Allowed Trade Unit	0.5 g (minimum)	System enforced	Users cannot sell or trade partial grams; all transactions must be in 0.5 g increments.
Allowed Physical Gold	5.0 g (minimum)	Enforced	

Derived Implementation Rules

- Fee Calculation Order
 - For sell/exchange:
 - $\text{net_amount} = (\text{grams} * \text{current_rate}) * (1 - \text{fee}\%)$
 - For buy:
 - $\text{grams} = (\text{amount} / \text{current_rate}) * (1 - \text{gateway_fee}\%)$
 - All fees recorded in the transaction_fee column.
- Approval Matrix
 - admin_required = true for any action involving store/bank.
 - Auto-approved only for in-app buys and gamified events.
- Audit & Ledger
 - Every fee applied generates a ledger entry:
 - Transaction → Fee → Destination Account
 - All manual actions (store buys, exchanges) log admin_id.
- Constraints
 - Minimum trade size = 0.5 g.
 - Wallet balance must remain ≥ 0 g.
 - Pending bank sell cannot overlap with other active sell orders.

Not in scope** Gamified Level Rules (ex:)

Level	Gram Threshold	Symbolic Meaning
Nosepin	2.5g	Entry milestone
Ring	5.0g	First savings goal
Churi	10.0g	Mid-tier achievement
Chain	25.0g	Premium milestone

User Q/A

User Research and Testing Results

Users:

What is Pakashona:

Pakashona মূলত একটি unified platform তৈরি করছে, যেখানে বাংলাদেশে প্রচলিত ২৪K Taati Bazaar gold rate এক জায়গায় দেখা যাবে। এই রেট বাজারের বিভিন্ন Taati Bazaar সোর্স থেকে নেওয়া বাস্তব মার্কেট রেফারেন্স, যাতে মানুষ সহজে বুঝতে পারে আজ সবচেয়ে ভালো এবং সস্তা ২৪K সোনার দাম কত।

এই প্ল্যাটফর্মে আপনি চাইলে ২৪K pure gold ডিজিটালি কিনে ধরে রাখতে পারবেন। এটি আপনার সেভ করা ভ্যালু সরাসরি সোনার দামের সাথে যুক্ত থাকে।

যখন আপনি ফিজিক্যাল গোল্ড নিতে চাইবেন, তখন আপনার ২৪K ডিজিটাল ভ্যালু অনুযায়ী ২২K বা ২১K সোনার সমপরিমাণ ওজন দেওয়া হবে। শুধু জুয়েলারি making charge এবং নির্দিষ্ট service fee প্রযোজ্য হবে, যা আগেই পরিষ্কারভাবে জানানো থাকবে।

Pakashona থেকে নেওয়া সব ফিজিক্যাল গোল্ড হবে হলমার্ক করা এবং অথেনটিক, এবং তা পাওয়া যাবে আমাদের যাচাইকৃত জুয়েলারি পার্টনারদের মাধ্যমে। আমাদের current vault ও jewellery partner হল I.K. Jewellers Ltd.

Short Answer:

Pakashona মূলত একটি unified platform তৈরি করছে, যেখানে বাংলাদেশে প্রচলিত ২৪K Taati Bazaar gold rate এক জায়গায় দেখা যাবে। আমরা বাজারভিত্তিক ২৪K gold price রেফারেন্স দেখাই এবং ভবিষ্যতে সেই ভ্যালু অনুযায়ী 22K/21K ফিজিক্যাল গোল্ড রিডেম্পশনের সুবিধা দিচ্ছি।

Pakashona থেকে নেওয়া ফিজিক্যাল গোল্ড হবে হলমার্ক করা ও সম্পূর্ণ অথেনটিক, এবং তা পাওয়া যাবে আমাদের যাচাইকৃত জুয়েলারি পার্টনারদের মাধ্যমে।

আমাদের বর্তমান vault ও jewellery partner হল I.K. Jewellers Ltd.

Out of Scope

Future Enhancements (Post-POC)

1. **Payment Gateway Integration (bKash/Nagad).**
2. **Vault Data Subscription Tier.**
3. **Multi-Store Expansion.**
4. **Referral Rewards.**
5. **Gift/Family Accounts.**
6. **Real-time gold price API integration.**

P2P 24K Marketplace (Premium Subscription Feature)

The goal is to create a secure, user-friendly marketplace for peer-to-peer trading, with final settlement happening physically at a partner store. The interface should feel familiar, like Facebook Marketplace.

- Vault data visibility or subscriptions.
- Multi-store setup.
- Real-time price API (admin manually inputs price).
- Family/gifting features.
- Fraud or code forgery validation.
- iOS/Android build (MVP stage).